Tô Gia Bảo

Data Engineer

Thu Duc, Ho Chi Minh City, Vietnam

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Summary

Data Engineer building Python and SQL pipelines, orchestration workflows, cloud data platforms, and conversational analytics tools. At Timo Digital Bank, delivered MCP-powered Slack integrations, operational automation, and analytics solutions with dbt and Lightdash. Hands-on with Airflow, Prefect, Amazon Athena, Amazon S3, PostgreSQL, MongoDB, Docker, and modern AI and lakehouse technologies, with a focus on reliable, maintainable, analytics-ready systems.

Experience

Timo Digital Bank by BVBank — *Ho Chi Minh City, Vietnam*

Data Engineer — *Aug. 2025 – Present*

- Fulfilled on-demand data requests by using Amazon EMR and AWS Glue to sync MariaDB data from the Timo platform into the landing layer, applying PII masking based on field sensitivity before the curated layer, and using dbt to populate the analytics layer.
- Implemented a FastMCP server for Slack, grounding conversational analytics in 3 context sources: dbt docs, Lightdash dashboards, and semantic-layer metadata.
- Replaced 2 recurring manual checks for MongoDB backup validation and general ledger lookups with automated workflows.
- Developed GenBI dashboards with Lightdash, dbt, and Airflow to monitor 3 operational areas: DAG health, Athena query performance, and Amazon S3 usage.
- Built Tracking247 for the Customer team, accepting tracking codes in Streamlit and using Airflow to sync shipment status to MongoDB, centralizing updates and reducing manual follow-up.

Projects

VN Market Pulse — *GitHub*

AI Research & Content Platform — *Jul. 2026*

- Built a Vietnamese web-research pipeline that plans LLM queries, searches across Brave, Tavily, Serper, and Exa, deduplicates and reranks sources, extracts bounded context, and generates source-backed Facebook posts.
- Integrated PydanticAI with an OpenAI-compatible Timo gateway for typed output, validation, and usage tracking; added retrieval budgets, retries, and CLI and Streamlit interfaces.

Mini Lakehouse — *GitHub*

Lakehouse & Data Platform — *Jul. 2026*

- Designed a lakehouse that provisions Amazon S3 storage and IAM access with Terraform, using Apache Iceberg, Apache Polaris, Trino, dbt, and Prefect across landing, curated, and analytics layers.
- Engineered contract-driven GitHub Archive and arXiv pipelines with idempotent Iceberg writes; defined catalog, RBAC, and maintenance policies; added dbt tests, documentation, and GLM-OCR on Kaggle GPUs.

Technical Skills

Programming	Python, SQL, C++
Data Engineering	Airflow, Prefect, Dagster, dbt, PySpark, Polars
Cloud & Infrastructure	Amazon S3, Amazon Athena, Amazon EMR, AWS Glue, Terraform, Docker, Git
Lakehouse & Databases	Apache Iceberg, Apache Polaris, Trino, PostgreSQL, MariaDB, MongoDB
AI & Analytics	Model Context Protocol (MCP), FastMCP, PydanticAI, semantic layer design, Lightdash, Streamlit, Slack

Education

University of Science, VNU-HCM (HCMUS) — *Ho Chi Minh City, Vietnam*

Bachelor of Data Science | GPA: 3.78/4.00 — *2022 – 2026*

Awards & Certifications

2025 **Top 25**, AIC Challenge
2025 **Recipient**, Ninety Eight Scholarship
2024 **Top 10**, RMIT Data Analytics Competition
2024 **Google Business Intelligence**, Professional Certificate
2024 **Google Data Analytics**, Professional Certificate

Publication

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval — *SOICT*

Co-author — *2025*

Languages

English TOEIC Listening & Reading: 910 | TOEIC Speaking & Writing: 350